

Lecture#11

Poisson Distribution

Let λ be the mean of successes in a specified time or space and the random variable X is the number of successes in a given time interval or specified region. Then X follows Poisson distribution as ($X \sim \text{Poisson}(\lambda)$)

$$f(x) = \frac{e^{-\lambda} \lambda^x}{x!}, \quad x = 0, 1, \dots, \infty$$

where $e = 2.718$.

□ Poisson distribution is used for count data. [$\lambda = \text{mean} = \text{variance}$].

Prove that the mean and variance of a Poisson distribution are equal.

(Note. $\sum_{y=0}^{\infty} \frac{\lambda^y}{y!} = 1 + \lambda + \frac{\lambda^2}{2!} + \frac{\lambda^3}{3!} + \dots = e^{\lambda}$)

Note. For variance calculation, first calculate $E[X(X-1)]$.

Example: The average number of calls received by a telephone operator during a time interval of 10 minutes during 5PM to 5:10 PM daily is 3. What is the prob. that the operator will receive

- i. no call
- ii. exactly one call
- iii. at least two calls tomorrow during the same time interval.

Solution: Let X be the random variable representing the number of calls made during the interval.

$$X \sim \text{Poisson}(3),$$

$$f(X=x) = \frac{e^{-3} 3^x}{x!}, \quad x \geq 0$$

$$i. P(X=0) = \frac{e^{-3} 3^0}{0!} = 0.0498$$

$$ii. P(X=1) = \frac{e^{-3} \cdot 3}{1!} = 0.1494$$

$$iii. P(X \geq 2) = 1 - P(X=0) - P(X=1)$$

$$= 1 - 0.0498 - 0.1494 \\ = 0.8008$$

Mean:

$$E(X) = \mu = \lambda$$

Variance:

$$V(X) = \mu = \lambda$$

Example: Find the mean and standard deviation of a Poisson variate X for which $P(X=1) = P(X=2)$.

Solution: Let $X \sim \text{Poisson}(\lambda)$

$$P(X = 1) = \frac{e^{-\lambda} \cdot \lambda^1}{1!} \\ P(X = 2) = \frac{e^{-\lambda} \cdot \lambda^2}{2!} \\ \therefore \frac{e^{-\lambda} \cdot \lambda^1}{1!} = \frac{e^{-\lambda} \cdot \lambda^2}{2!} \\ \Rightarrow \lambda = \mu = 2$$

So, the mean = 2, Standard deviation = $\sqrt{2}$.

Exercise.

1. A call center receives an average of 4 calls per hour.
 - (a) What is the probability that exactly 7 calls are received in an hour?
 - (b) What is the probability that no calls are received in an hour?
2. Suppose accidents occur at a busy intersection at an average rate of 2 per week.
 - (a) What is the probability that at most 1 accident occurs in a week?
 - (b) What is the probability that at least 2 accidents occur?
3. A call center receives an average of 6 calls per hour.
 - (a) Find the probability that the call center receives between 4 and 8 calls inclusive.

(b) Find the probability that the call center receives more than 10 calls.

Poisson approximation to binomial distribution:

If n becomes larger and p becomes smaller, then the Poisson distribution with $\mu = \lambda = np$ provides an approximation to the binomial distribution.

Example: The probability of breaking a glass beaker while heating in the laboratory is 0.012. If we heat 250 such glass beakers, find the probability that there will be exactly 2 breaks.

Solution: Let X be the number of breakages

$X \sim \text{Binomial}(n = 250, p = 0.012)$

$$\text{Now, } P(X = 2) = \binom{250}{2} 0.012^2 (1 - 0.012)^{250-2} = 0.2245$$

Since n is large and p is small, we may obtain an approximate probability by using the Poisson distribution with $\mu = \lambda = np = 250 \times 0.012 = 3$.

$$P(X = 2) = \frac{e^{-3} \cdot 3^2}{2!} = 0.2241.$$

Example: In a certain industrial facility, accidents occur infrequently. It is known that the probability of an accident on any given day is 0.005 and accidents are independent of each other.

a. What is the probability that in any given period of 400 days there will be an accident on one day?

b. What is the probability that there are at most three days with an accident?

Solution: Let X be a binomial random variable with $n = 400$ and $p = 0.005$.

Thus, $np = 2$.

Using the Poisson approximation,

$$(a) P(X = 1) = \frac{e^{-2} \cdot 2^1}{1!} = 0.271$$

$$\text{and (b) } P(X \leq 3) = P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3)$$

$$= \frac{e^{-2} \cdot 2^0}{0!} + \frac{e^{-2} \cdot 2^1}{1!} + \dots = 0.857$$

Example: In a manufacturing process where glass products are made, defects or bubbles occur, occasionally rendering the piece undesirable for marketing. It is known that, on average, 1 in every 1000 of these items produced has one or more bubbles. What is the probability that a random sample of 8000 will yield fewer than 7 items possessing bubbles?

Solution: This is essentially a binomial experiment with $n = 8000$ and $p = 0.001$. Since p is very close to 0 and n is quite large, we shall approximate with the Poisson distribution using

Mean, $\lambda = (8000)(0.001) = 8$.

Let, X represents the number of bubbles, we have

$$P(X < 7) = \sum_{x=0}^6 \frac{e^{-\lambda} \lambda^x}{x!} = 0.3134.$$

Exercise.

1. A factory produces 1000 items per day. Each item independently has a probability 0.002 of being defective. Find-
 - (a) Probability exactly 3 defective items
 - (b) Probability at most 3 defective items
 - (c) Probability at least 3 defective items
2. A store has 500 customers in a day. Each customer independently has a 0.01 probability of returning a product. Let X = number of returns. Calculate
 - (a) $P(X = 4)$
 - (b) $P(X \leq 4)$
 - (c) $P(X \geq 6)$
3. A drug has a 0.005 chance of causing a rare side effect. 600 patients take the drug. Let XXX = number of patients with side effect.
 - (a) $P(X = 2)$
 - (b) $P(X \leq 2)$
 - (c) $P(X \geq 4)$
4. A server receives 2000 emails per day. Each has 0.1% chance of being spam. Let XXX = number of spam emails.
 - (a) $P(X = 3)$

(b) $P(X \leq 3)$

(c) $P(X \geq 3)$

5. A factory produces 8000 light bulbs. Each has 0.05% chance of being defective. Let XXX = defective bulbs.

(a) $P(X = 2)$

(b) $P(X \leq 2)$

(c) $P(X \geq 3)$